

BOUROUBA ASMA

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EDUCATION

State Engineer's Degree and Master 2 in Computer Science

2019 – 2024

Specialization: Artificial Intelligence and Data Science

École Supérieure en Sciences et Technologies de l'Informatique et du Numérique (ESTIN)

Amizour, Béjaïa, Algeria

EXPERIENCE

Wedey

September 2025 - March 2026

AI & Automation Engineer

Algeria (Remote)

- Built and deployed **n8n-based automation** workflows integrating databases like **Supabase** and SaaS tools via **APIs and webhooks** to eliminate manual tasks and ensure automatic data flow across systems.
- Designed **AI agents** with both autonomous and human-in-the-loop execution modes, applying **prompt engineering** techniques to optimize agent behavior and reliability in production.
- Evaluated business processes to identify automation opportunities, then tested and debugged deployed workflows to ensure failure handling and operational stability.
- Built user-facing interfaces with **Bubble** to allow business users to interact with and manage deployed automation solutions.

Bastyon AI

March 2024 - September 2024

LLM Engineer

Algeria (Remote)

- Built a **RAG** system using **LlamaIndex** with **Qdrant**, connecting private document collections to LLMs from Hugging Face and OpenAI, working within a small engineering team.
- Experimented with chunking and indexing strategies, retrieval tuning, and prompt templating to iteratively optimize retrieval quality and generation accuracy.

Cevital Group

January 2024 - June 2024

AI & Data Science Intern

Bejaia, Algeria

- Designed a **predictive maintenance** system from textual intervention reports, combining a BERTopic clustering model with a Bayesian inference layer to predict next interventions and optimize maintenance scheduling.
- Built the full pipeline in **Python**, from data preprocessing through model evaluation.
- Presented project results in a paper titled: *Advanced Analytics for Predictive Maintenance: A Case Study Using BERTopic and Bayesian Modeling*, submitted to Springer, highlighting the approach used and the performance achieved.

Numilex

September 2024 - June 2025

Web Developer

Bejaia, Algeria

- Created responsive, user-friendly, and high-performance web interfaces based on client requirements.
- Used HTML, CSS, Bootstrap, and JavaScript to translate functional requirements into concrete solutions.

Entreprise Portuaire de Béjaïa - EPB

August 2022 - March 2023

Web Development Intern

Bejaia, Algeria

- Designed an end-to-end staff on-call management application.
- Performed UML modeling and database structuring with PhpMyAdmin.
- Developed the web interface with HTML, CSS, Bootstrap, and PHP to automate a manual process.

TECHNICAL SKILLS

AI, Data Science & LLM

- **Languages:** Python, Matlab, SQL
- **Python Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **NLP:** NLTK, SpaCy, BERTopic
- **Automation Tools:** n8n, Botpress
- **LLM & Integration:** OpenAI API, Hugging Face, LangChain, LangGraph, LlamaIndex, Haystack, PandasAI
- **Computer Vision:** YOLOv8, Roboflow
- **Vector Databases:** Qdrant, Supabase
- **Analytics:** eViews

Software & Web Development

- **Languages:** C, Dart, JavaScript
- **Backend & APIs:** FastAPI
- **Web:** HTML, CSS, Bootstrap
- **Databases:** MySQL, PostgreSQL, MongoDB, SSMS
- **Tools:** GitHub, VS Code, Anaconda, Bubble.io
- **Methodologies:** Scrum, Kanban
- **Design:** Figma, Canva

KEY KNOWLEDGE

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|--------------------------------------|-------------------------|
| • Natural Language Processing | • Deep Learning |
| • Large Language Models (LLM) | • Data Science |
| • Generative Artificial Intelligence | • Data Analysis |
| • Machine Learning | • Business Intelligence |

LANGUAGES

Arabic	Native / Bilingual
English	Fluent
French	Intermediate